Technical Analysis of the k-Nearest Neighbors Algorithm for Identifying Species of Iris from the Fisher Iris Dataset

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***Abstract*— This report is a technical analysis of the K-Nearest Neighbors (kNN) algorithm when used in supervised machine learning to analyze the Fisher Iris data. Our modular code system classifies morphmetric features of iris species (according to their morphometrics) with an accuracy of 96.67% at k = 5. Besides the laboratory work, we have learned about real-life applications of kNN to government tasks: identifying Medicare/CMS healthcare fraud, zoning land use in Cape Town, South Africa, and disaster response requirements forecasting in Jakarta, Indonesia. Through trial and error and confusion matrices, we discovered that kNN works best when data is separated linearly but fails at the boundaries of classes. The comprehensive audit of flood-prediction models revealed that preprocessing and hyper-parameter adjustments should be performed with great care. All in all, the findings suggest that kNN needs substantial data preparation with high-dimensional data, but due to its transparency, kNN is more ethical than opaque neural networks when it comes to making decisions about the public administration that should be administered under the law with references to legal due process.**

***Keywords*— k-nearest neighbors (knn), supervised machine learning, fisher iris data, morphometric features, accuracy, healthcare fraud, land use zoning, disaster response forecasting, confusion matrices, data separation, flood prediction models, data preparation, high-dimensional data, transparency, public administration**

I. Introduction

The k -Nearest Neighbors (kNN) algorithm is still a pillar of supervised machine learning due to its lax architecture of lazy learning. Rather than generalization being done by fitting a model during training, kNN delays generalization until a query is made. Although current AI studies have moved to opaque deep neural networks, kNN remains of paramount importance to the public sector, where it is required by law to have transparency in its algorithm.

This paper will aim to develop an automated classification system that will allocate the iris flowers to either one of three species, including Iris setosa, Iris versicolor, and Iris virginica, based on four quantitative data, including sepal length, sepal width, petal length, and petal width.

The dataset to be used is the classical Fisher Iris dataset, which was published by Ronald Fisher in 1936 as a part of his original research on the discriminant analysis topic. It has 150 samples, 50 samples of each species, and is a multi-class classification problem of different difficulty: one of the classes can be linearly separated from the rest, and the other two are heavily overlapped in the distribution of their features.

The kNN algorithm is based on a very simple geometric concept: related data points are clustered more closely together in feature space. Where an unidentified sample is introduced, kNN compares the distance of the unknown sample to each training set point, and the k closest points are identified, and the majority vote among the k nearest points to the sample is the class label.

The purpose of this report is twofold: to capture the technical application of kNN to the Iris dataset and offer a methodical performance analysis; and secondly, to explore the experience of kNN implementation in operational use in the real-world examples of government activities published in the past five years (2021-2026).

II. Methodology

A. Modular Code Architecture

To ensure reproducibility and maintainability, I structured the code into six logical modules, which play a distinct role in the machine learning workflow.

1) Module 1: Library Imports: The first module establishes all dependencies required for the program. This prevents runtime errors by ensuring all necessary tools are available before execution begins.

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import
    train_test_split
from sklearn.neighbors import
    KNeighborsClassifier
from sklearn.metrics import accuracy_score,
    classification_report, confusion_matrix
```

The imports include matplotlib for visualization, sklearn components for machine learning functionality, and metrics for performance evaluation.

2) Module 2: Data Acquisition: The second module loads the Iris dataset and separates it into features (X) and target labels (y).

```
data = load_iris()
X = data.data       # Feature matrix: shape
    (150, 4)
y = data.target     # Target vector: shape
    (150,)
```

This distinction is crucial to supervised learning: X will contain the measurements of the input, whereas y will contain the correct class labels.

3) Module 3: Data Splitting: The third module partitions the data into training and testing subsets using an 80/20 split.

```
X_Train, X_Test, y_Train, y_Test =
    train_test_split(
    X, y, test_size=0.2, random_state=0
)
```

To prevent measuring memorization exclusively, an 80/20 split is used where the model is trained on 120 samples and tested on 30 randomly chosen examples that are not visible. The initialization of random state=0 ensures that the split is reproducible.

4) Module 4: Model Training: The fourth module instantiates and trains the kNN classifier.

```
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_Train, y_Train)
predictions = knn.predict(X_Test)
```

The `n_neighbors=5` parameter configures the voting mechanism to use 5 nearest neighbors. Unlike parametric models, the `fit()` method does not perform mathematical optimization; it simply stores the training data in memory. Computation is deferred until `predict()` is called.

5) Module 5: Performance Evaluation: The fifth module generates quantitative performance metrics.

```
accuracy = accuracy_score(y_Test, predictions)
print("Accuracy:", accuracy)
print("\nClassification␣Report:")
print(classification_report(y_Test,
    predictions))
cm = confusion_matrix(y_Test, predictions)
print("\nConfusion␣Matrix:")
print(cm)
```

This module calculates overall accuracy, per-class precision and recall, and generates the confusion matrix for detailed error analysis.

6) Module 6: Visualization: The sixth module creates visual representations of results.

```
plt.figure(figsize=(8, 6))
scatter = plt.scatter(X_Test[:, 0], X_Test[:,
    1],
                      c=predictions, cmap='
                          viridis', alpha=0.8)
plt.xlabel(data.feature_names[0])
plt.ylabel(data.feature_names[1])
plt.colorbar(scatter, label='Predicted␣Class')
plt.show()
```

```
plt.figure(figsize=(6, 6))
plt.imshow(cm, cmap='Blues')
plt.colorbar()
plt.xticks(ticks=range(3), labels=data.
    target_names)
plt.yticks(ticks=range(3), labels=data.
    target_names)
plt.show()
```

A scatter plot provides the distribution of the predictions in feature space, whereas a heatmap of the confusion matrix is an intuitive display of the error of classification.

B. Rationale for Modular Design

This modular structure follows professional software engineering practices. Each module has a single, well-defined responsibility, making the code easier to debug, test, and modify. For example, if preprocessing steps needed to be added, they could be inserted as a new module between data acquisition and splitting without affecting other components. This separation of concerns ensures that changes to one module do not create cascading failures in others.

III. Results and Experimental Analysis

A. Baseline Performance Results

The kNN model configured with $k = 5$ achieved an overall accuracy score of **96.67%**, correctly classifying 29 out of 30 test samples on the specific random split generated by `random_state=0`.

The detailed classification report revealed the following per-class performance:

- **Setosa (Class 0):** Precision = 1.00, Recall = 1.00, F1-Score = 1.00, Support = 11
- **Versicolor (Class 1):** Precision = 1.00, Recall = 0.92, F1-Score = 0.96, Support = 13
- **Virginica (Class 2):** Precision = 0.86, Recall = 1.00, F1-Score = 0.92, Support = 6

These measures show that Setosa was rated exactly, while Versicolor and Virginica had small inaccuracies.

B. Hyperparameter Experimentation

In order to explore the performance with respect to the k parameter, I set n neighbors to different values (ranging between 1 and 20) and measured the resulting accuracy on the same test set.

1) Low k Values ($k = 1$ to $k = 3$): Accuracy ranged from 96% to 98%. In the case of k=1, the model is hyper-sensitive to each training sample, and the irregular decision boundaries follow all the outliers. This represents the classic symptom of overfitting, where the model memorizes specific training examples rather than learning generalizable patterns.

2) Optimal Range ($k = 5$ to $k = 10$): Accuracy remained consistently between 96% and 97%. This range represented a balance between local sensitivity and global stability. The model reduces random noise when there are 5-10 neighbors who vote, but still the desired operating point on the bias-variance curve can be detected, i.e. local class boundaries.

3) High k Values ($k = 15$ to $k = 20$): Accuracy declined to 93–94%. With 20 neighbors in the vote, the global pattern of classes dominates the local pattern of classes. In case there are more Versicolor in the training set on the whole, the model can label Versicolor when it is a definite Virginica cluster, demonstrating underfitting: the model is too simple to represent the complexity of the actual data.

The experimentation confirmed that $k = 5$ provides optimal performance for this dataset, balancing bias and variance effectively.

C. Confusion Matrix Analysis

The confusion matrix generated for the $k = 5$ configuration revealed the following structure:

	Predicted		
Actual	Setosa	Versicolor	Virginica
Setosa	11	0	0
Versicolor	0	12	1
Virginica	0	0	6

1) Perfect Setosa Identification: There were zero errors with all the 11 Setosa samples. This confirms that Setosa is geometrically different in feature space, and it is a compact, isolated cluster that is distinctly separated by the other two species. This segregation is due to the fact that the petal measurements of Setosa are significantly lower which results in a natural segregation in the 4-dimensional feature space.

2) Boundary Confusion Between Versicolor and Virginica: There is just one off-diagonal element in the matrix: there was a single Versicolor sample that was confused with Virginica (Row 1, Column 2). This mistake is indicative of the biological fact that these two species are phenotypically similar whose morphometric ranges overlap.

3) Diagnostic Interpretation: The confusion matrix is accurate in the sense that it provides an accurate account of model behavior: kNN is good at the classes that are linearly separable (Setosa) and predictable problems in the boundary areas where biological overlap forms ambiguous features distributions (Versicolor vs. Virginica). The wrongly grouped Versicolor sample must have been in an overlap zone, with 5 of its closest neighbors being mostly Virginica samples, and therefore, the voting mechanism reached a false conclusion.

This trend is not an algorithm failure but rather the manifestation of natural uncertainty of the underlying data. The confusion matrix has been able to show where there would be most human expert review value.

IV. Real-World Applications

A. Real-World Government Applications

Upon the analysis of the published studies since 2021 and 2026, I identified three key areas in which kNN and other distance-based algorithms are applied in governmental tasks.

1) Healthcare Fraud Detection (Medicare/CMS, United States): Tajrobehkar et al. (2024) studied the use of machine-learning to identify fraud in Medicare claims [4]. They examined utilization patterns in order to indicate fraudulent billing practices. Farahmandazad et al. (2025) discussed the issues of the class-imbalance in Medicare fraud detection, stating that fraud cases represent a minor section of all possible claims and need special methods to be classified successfully [5].

In this area, predictions are dichotomous: a statement can be either fraudulent or valid. The distance-based techniques can identify an outlier among provider billing patterns. Those providers whose billing is significantly out of line with their peers are spotted.

2) Urban Land-Use Zoning (Cape Town, South Africa): Lefulebe et al. (2023) used K-Nearest Neighbour classification on the Satellite imagery of the Cape Flats area in Cape Town, South Africa [6]. PlanetScope satellite images were used to classify land use and cover into categories, which include built-up areas, vegetation, water bodies, and trees. According to the study, there was 96.54 per cent accuracy in this multi-class task, and this indicated the effectiveness of KNN in identifying individual spectral signatures with satellite data [6].

3) Disaster Response Prediction (Jakarta, Indonesia): K-Nearest Neighbors with machine-learning optimization were used to design a flood-risk forecasting system with Jakarta Province [7]. The system forecasts the level of risk of floods depending on the environmental and meteorological parameters. In Lumbantobing et al. (2024), KNN was implemented to predict flood in Jakarta based on the weather information through rainfall, which can then be used to predict floods [8].

The categorical predictions are used to group the administrative regions in terms of the risk levels (Low, Moderate, or High). The algorithm is an early-warning system, as it recognizes historical weather patterns that have the closest neighbors to the present weather.

B. Technical Specification Audit

To analyze the technical aspects, I chose the article by Ino and Taufik (2025), Flood Risk Prediction Using the K-Nearest Neighbors (KNN) method with machine learning optimization in Jakarta Province [7].

1) Distance Metric Selection: In cases where the environmental and meteorological data are continuous numerical values (like the intensity of rainfall and the height of water), Euclidean distance is a geometrically significant concept of similarity. Euclidean distance formula estimates the straightforward spatial scale in between the feature vectors:

$$d(p,q) = \sqrt{\sum_{i=1}^{n}(p_i - q_i)^2}$$

This measure is suitable for physical measurements where absolute numerical values are important. As an example, when the current conditions are recorded to have 150 mm of rainfall and a water level of 50 cm, and a past day was recorded to have had 148 mm of rainfall and a water level of 52 cm, the extent to which these two weather patterns resemble each other in a geometrically useful sense is measured by the Euclidean distance.

Alternative metrics would be less suitable:

- Manhattan Distance accumulates the absolute feature differences and does not take the geometry between features into account
- Cosine Similarity is used to measure angular relations and is a better technique to use in directional data or text

Where actual numerical data are physically significant in weather and floods, Euclidean distance is an appropriate measure of environmental similarity.

2) Preprocessing Requirements: Preprocessing is essential in kNN applications in which features have distinct values. The range of environmental variables is often extremely different:

- Rainfall: in millimeters (which may be 0-300mm or more)
- Water levels: in centimeters (which may be 0-100mm or more)
- Flood frequency: discrete counts (0-5 occasions)

In the absence of normalization, distance computations in a mathematical way are dominated by the features with higher numerical scales. Take two weather patterns in the past:

- Pattern A: Rainfall = 200mm, Water Level = 40cm, Frequency = 2
- Pattern B: Rainfall = 100mm, Water Level = 45cm, Frequency = 5

The difference in rainfall of 100mm would dwarf the difference in water level of 5cm when calculating Euclidean distance and make the smaller scale features insignificant.

Standard preprocessing approaches include:

- Min-Max Scaling: Transforms features to a fixed range (typically 0–1)
- Z-Score Normalization: Standardizes features to have mean=0 and standard deviation=1

The two approaches make sure that every feature has a constant contribution to the process of distance calculations, no matter what the measurement scale at the beginning.

3) Hyperparameter Selection: The selection of k represents a fundamental tradeoff in kNN modeling between bias and variance.

Low k values (e.g., $k = 1$) are very sensitive to local patterns but can be overfit by sensitivity to noise and outliers. The boundary of the decision becomes very irregular, and the decision is based on personal training samples.

High k values (e.g., $k = 20+$) provide stable predictions through averaging but risk underfitting by failing to capture local class structures. The decision boundary becomes overly smooth, potentially missing important distinctions.

The best k values are sensitive enough to local data, and stable enough to occur across different data partitions, and may largely be identified by systematic validation methods like cross-validation, where the different values of k are evaluated on various data splits to determine which value attains optimal performance on their validation data. The optimum k is too much data-dependent, and it must be empirically determined instead of utilizing default values.

C. Ethical Decision-Making

The application of kNN in life and death decisions in the government brings some serious concerns on the trade-off between interpretability and algorithm accuracy. The contemporary deep learning research can be described as computational black boxes that take inputs and give outputs with high accuracy, but internal decision-making mechanisms that are essentially opaque. A neural network may produce the result of 99% chance that this claim is fraudulent based on the patterns of millions of weight parameters, but the human brain can never identify the logical path between input and output.

Conversely, kNN is a glass box where all predictions are auditable by looking into the historical precedents which affected the decision. Take the actual case of the analysis of the confusion matrix in this paper: the model mistakenly labeled one Versicolor sample as Virginica. Since kNN is transparent, this error may be further investigated by making queries about the 5 nearest neighbors that resulted in the vote, which may disclose whether the misclassification was in a real boundary area where the biological overlap may occur.

This transparency carries great legal consequences in government business. In case Medicare refuses to reimburse a healthcare provider after the neural network has produced a probability result, the healthcare provider may appeal the ruling to due process since the agency cannot justify its decision by simply stating an opaque mathematical formula. Nevertheless, when the refusal is defined with the help of kNN analysis, the agency can provide concrete data by displaying the very historical cases that acted as close neighbors and leave the provider with the ability to analyze the precedents and comprehend the grounds behind the refusal. This is in line with the legal standard of precedent on which the administrative law is based.

This analysis suggests that kNN would be a suitable method to use when governmental decisions are high-stakes and transparency and accountability are legal necessities, even though the raw accuracy may be less than that of more complicated methods. Nevertheless, such credibility has significant conditions. To begin with, preprocessing is essential, as the Jakarta flood study proved that effective performance cannot occur without the appropriate feature scaling. Second, the issue of class imbalance should be resolved - the Medicare fraud detection study revealed that kNN cannot operate effectively with severely imbalanced data, unless it is combined with resampling methods. Third, there are opportunities and challenges with bias visibility: in the case of historical training data being biased, the bias will be reflected in the referred-to precedents. Transparency in kNN also means bias is audited (unlike in neural networks, where bias is obscured in abstract weight matrices), which is an advantage (easy to detect and correct) and also a weakness (easy to undermine system legitimacy).

kNN is an important standard of government AI: the open standard by which the mysterious black-box models are justified to be used. Where there is a right to explanation, e.g. healthcare, criminal justice, or social services, the interpretability of kNN gives the accountability demanded of due process and of trust in the Bureaucracy, despite more accurate yet opaque alternatives.

V. Acknowledgment

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VI. Conclusion

This paper has shown that the k-Nearest Neighbors algorithm is in a good position to offer high successful classification in the Iris dataset benchmark having a 96.67 percent accuracy at a given setting of k = 5. The systematic exploration of the values of k showed that hyperparameter tuning is critical to reducing bias and variance, and k = 5 gives the best results in this particular problem.

The confusion matrix analysis was diagnostic information (kNN is very effective at recognizing linearly separable classes (Setosa)) with predictable weaknesses in the gray area where biological overlap produces ambiguous feature distributions (Versicolor vs. Virginica). This trend is an inherent uncertainty of the data and not the failure of the algorithm.

Real-world government performance investigation re-discovered domain-dependent performance. The studies show that kNN is not able to perform well in the healthcare fraud detection task because of the significant class imbalance [5], but it is able to produce good outcomes in urban land-use prediction with satellite images [6], disaster response prediction [7], and [8] because of the favorable environmental conditions such as natural separation of classes, and cyclical phenomena.

The technical audit confirmed the imperativeness of the preproprocessing and hyper parameter optimization in operational deployment. The appropriate scaling of features makes all the measures be important in calculating distance whilst the systematics validation methods determine the best k values in selected datasets.

The usefulness of kNN in government processes is not its raw computation power-ensemble techniques often have greater accuracy-but rather its interpretability and accountability. The opportunity to trace predictions to particular historical precedents gives the transparency required to due process and public confidence in high-stakes areas in which administrative judgment is to be made on the rights and livelihood of the citizens [9]. The increased transparency of the algorithm renders the algorithm more ethical to use in the public sector where explainable AI is a legal requirement, acting as the transparent benchmark to which the opaque models should defend their use in government.
decision-making systems.

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